

COMPASS_Synoptic_LE_Soil-Fluxes_2023_CO2

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0.1 Goals of this code:

1. Reads in raw smart chamber
2. If no metadatafile, write one out & fields will be: Label the IDs pulled from the label, Plot *fixed so that if there are any issues in the label they are good here Drop column – false as long as label looks right adjust time – would be initially populated by the start time in the OR blank adjust observation length – initially populated by the obs length OR blank notes / comment
3. Code reads in the meta data and applies it to the data
 - a. i.e. drops fluxes that have the drop column TRUE or FLAG this
 - b. adjusting times as needed
4. Now you compute fluxes
5. Visualize and look for issues
6. If found, report to meta data file then go back to 3 and reanalyze
7. Apply algorithmic qaqc and generate flags

0.2 Set Up & Load Packages

0.3 Pull in Smart Chamber Files from the Raw Data folder

0.4 Unfortunately - we had many problems with labels - we need to include the sites in the Plot_ID

0.5 Create a unique Plot_ID, check if there is a meta-data file, if not, create a meta-data file

```
[1] "CSV file exists and has been read into the code."
```

0.6 Mark fluxes that were retaken - this is the first issue! Need to get all but last flux to be TRUE for drop - fieldnotes with explanation for flux retaken - usually weird zeros, Steven probe not inserted in the soil, or label mistake

##Dont need to pull L1 temp data or any other temps like we do for macrophyte because the smartchamber records the temp

0.7 Match up the metadata to the rest of the data

1 entry had no timestamp matches!

0.8 fixing some Deadbands

```
Warning in left_join(., time_overrides, by = "Plot_ID"): Detected an unexpected many-
to-many relationship between `x` and `y`.
i Row 30585 of `x` matches multiple rows in `y`.
i Row 16 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

```
##Extracting initial CO2 concentration
```

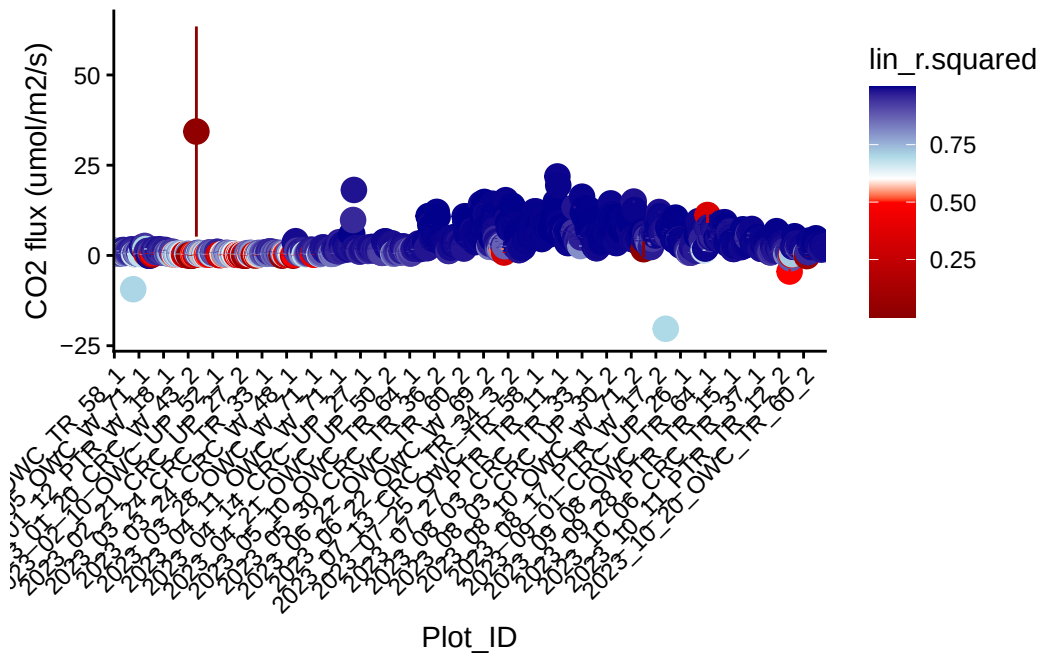
0.9 Change the units from ppb to nmol

Assuming atm = 101325 Pa

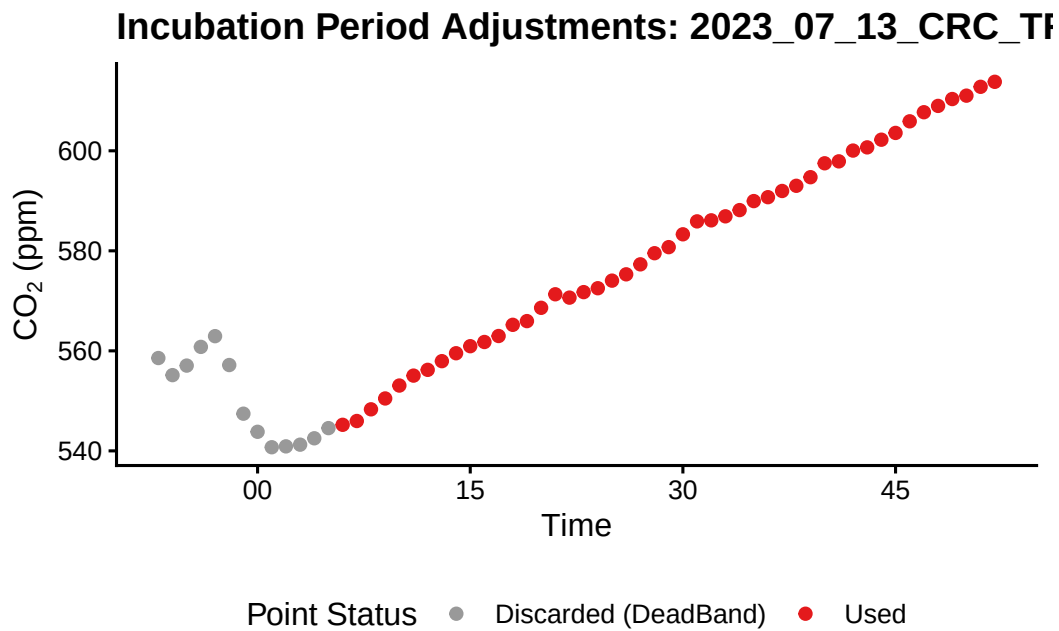
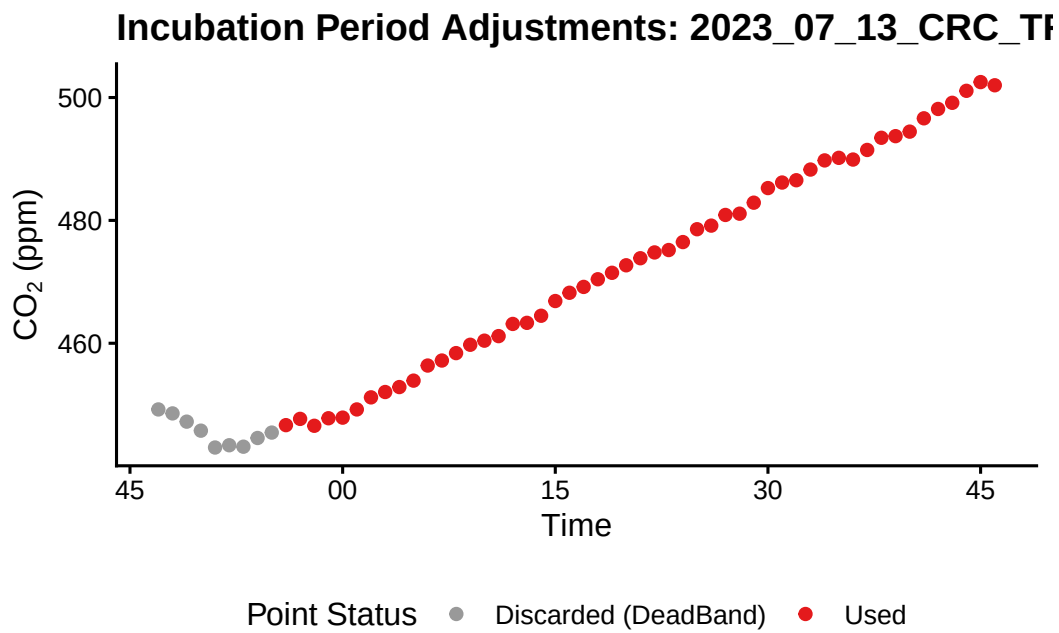
Using $R = 8.31446261815324 \text{ m}^3 \text{ Pa K}^{-1} \text{ mol}^{-1}$

0.10 Calculate fluxes

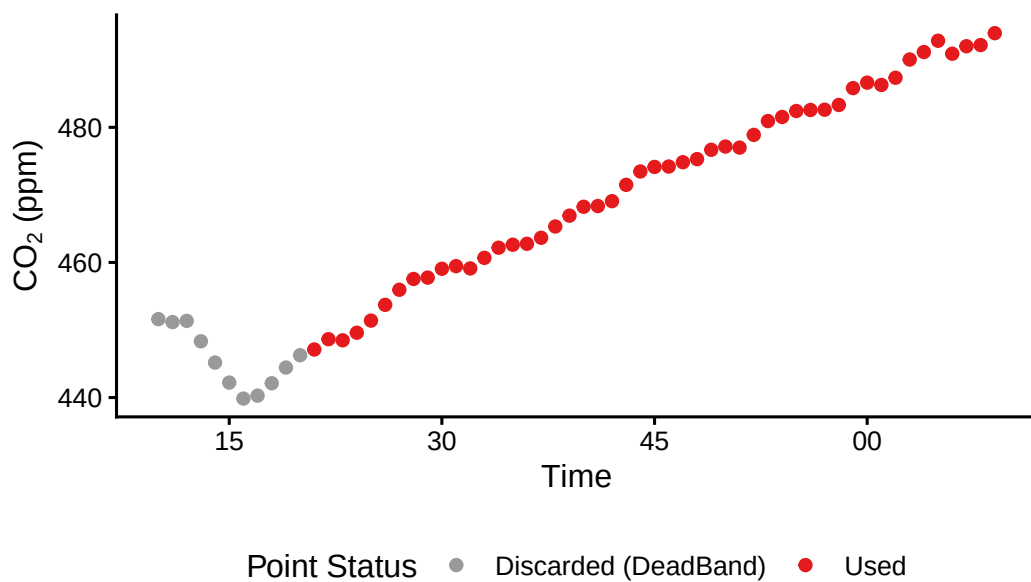
0.11 Visualize fluxes and sort by r2



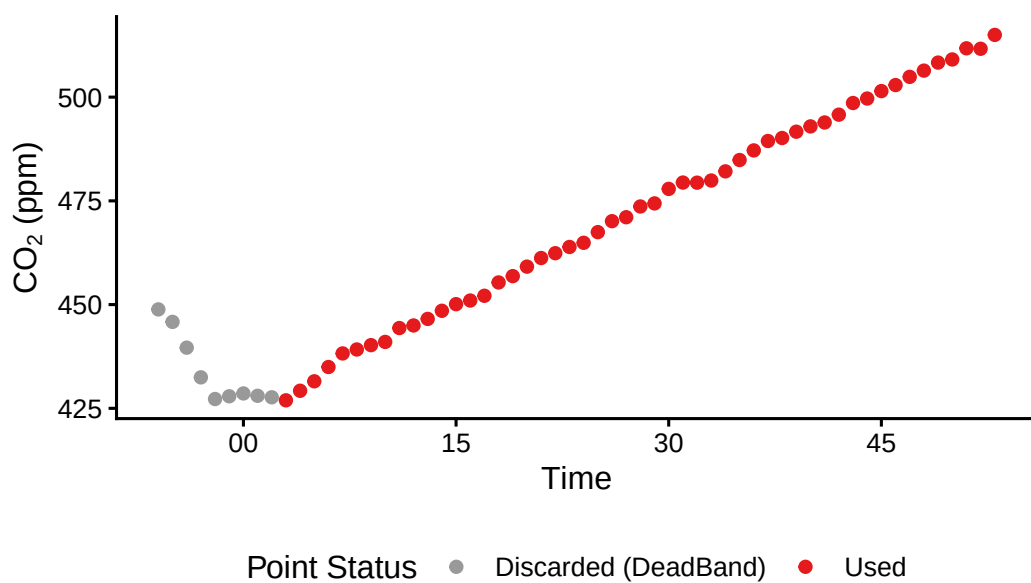
0.12 check the fluxes for which I changed the start time due to lags or ebullition (recorded in notes of metadata)

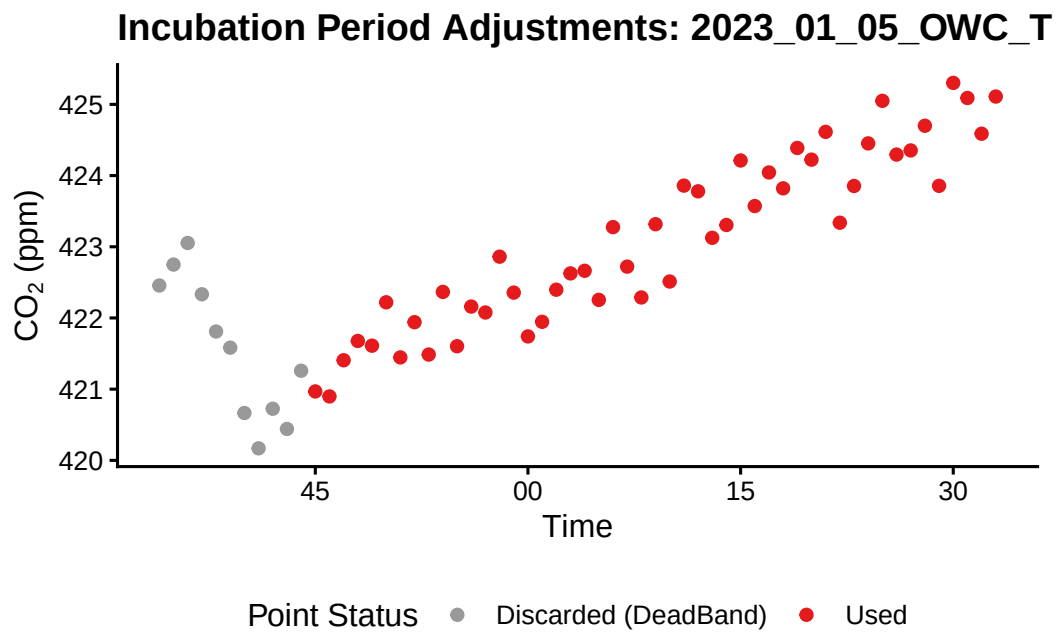
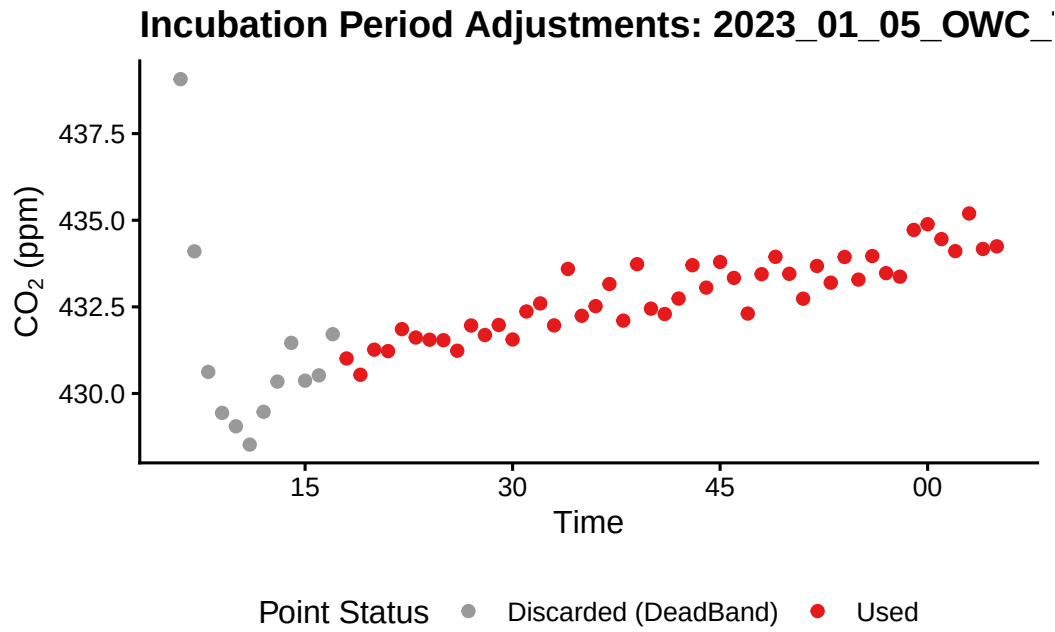


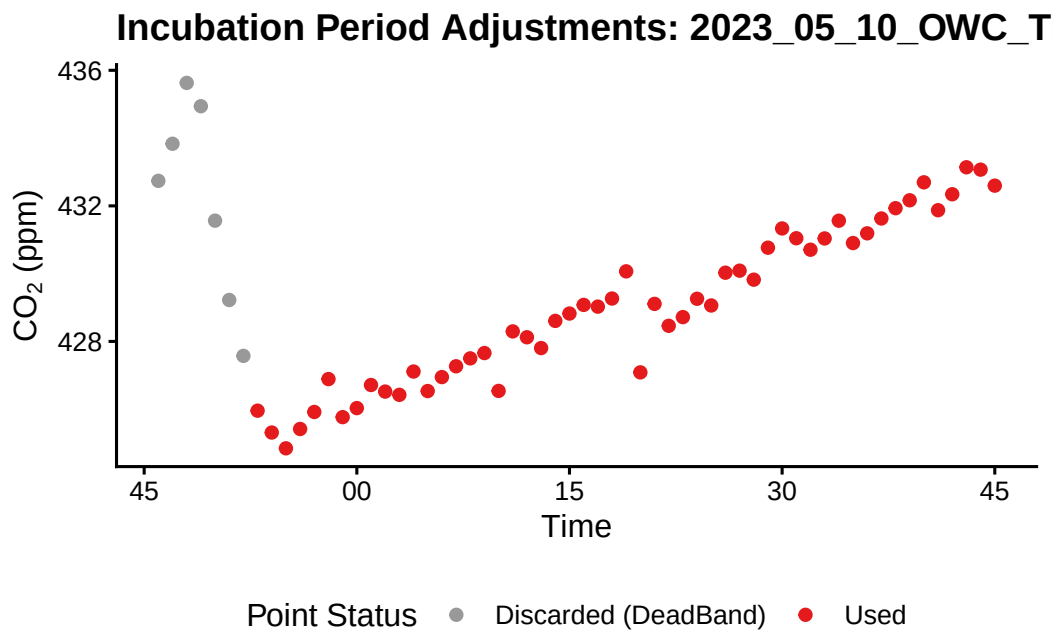
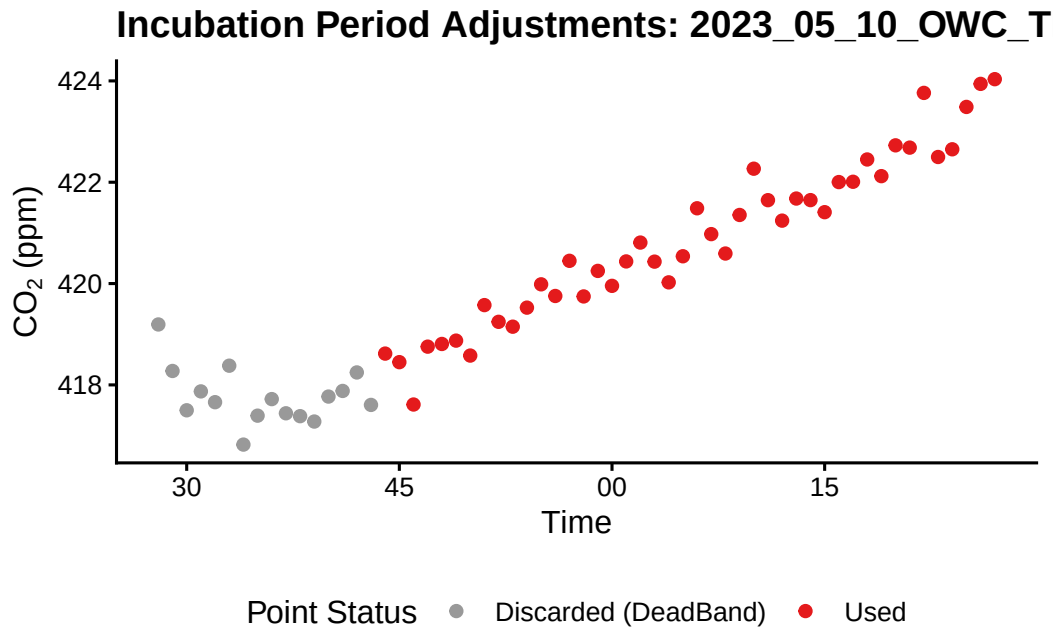
Incubation Period Adjustments: 2023_07_13_CRC_Tf



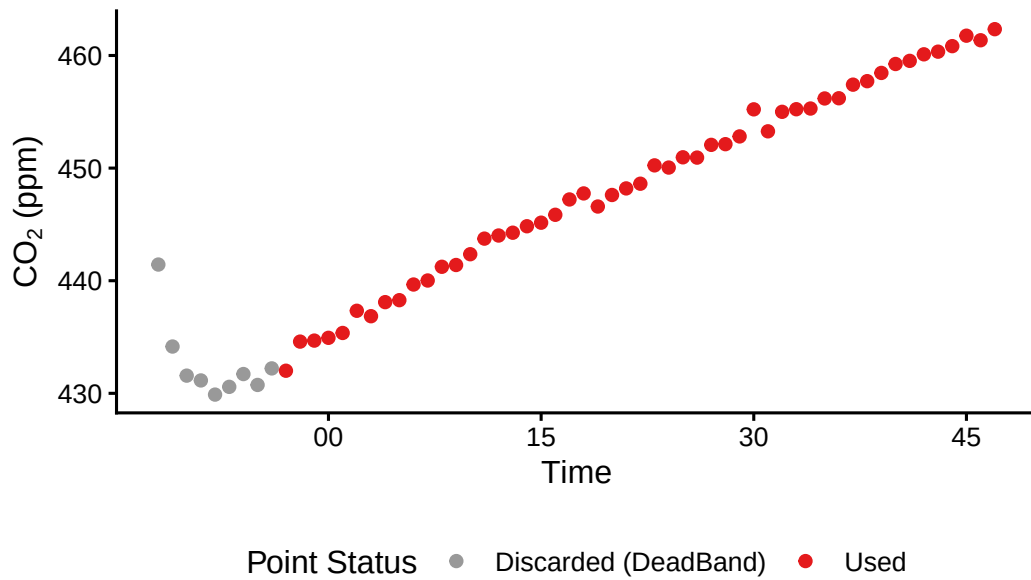
Incubation Period Adjustments: 2023_07_13_CRC_Tf



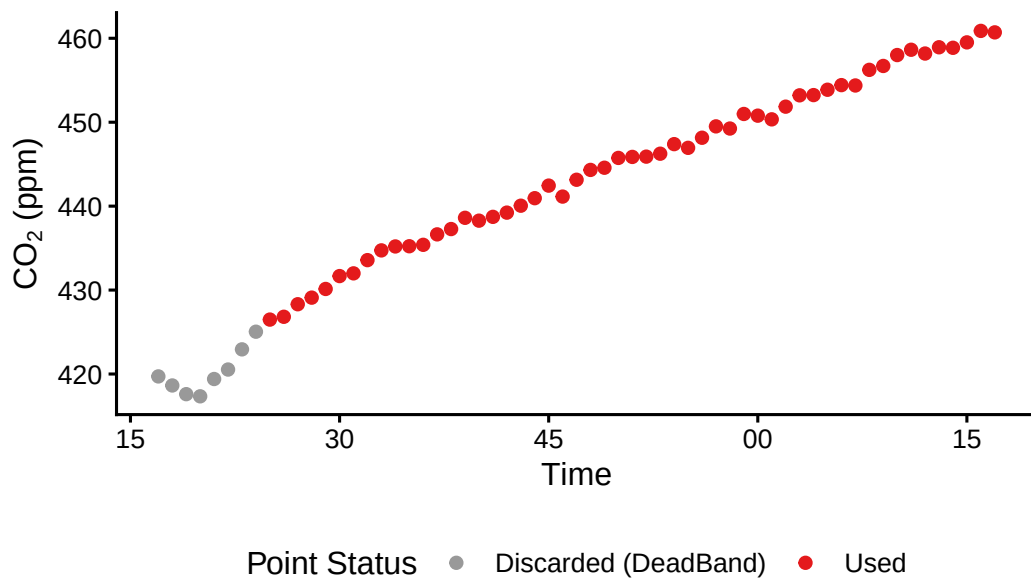




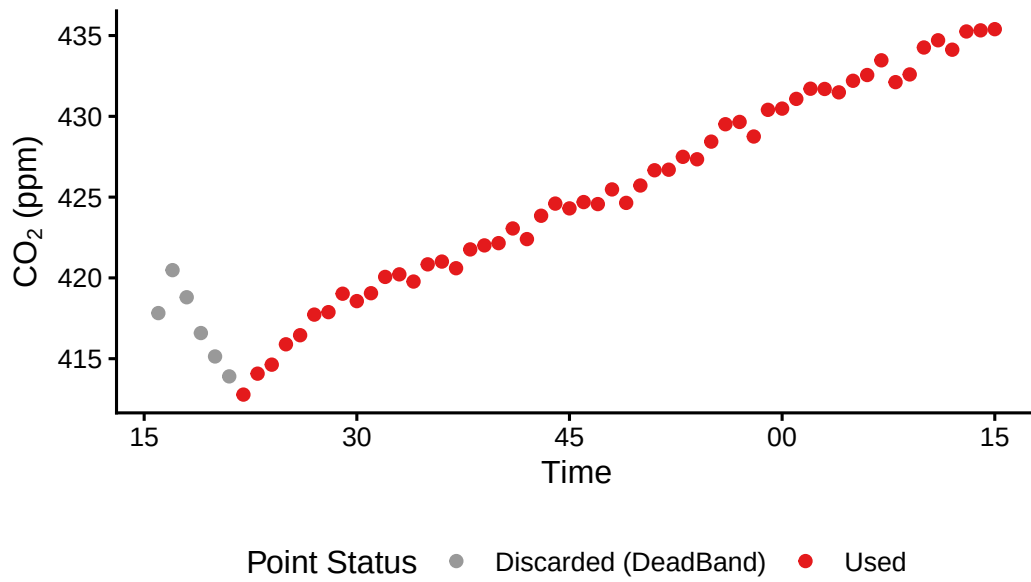
Incubation Period Adjustments: 2023_06_22_OWC_U



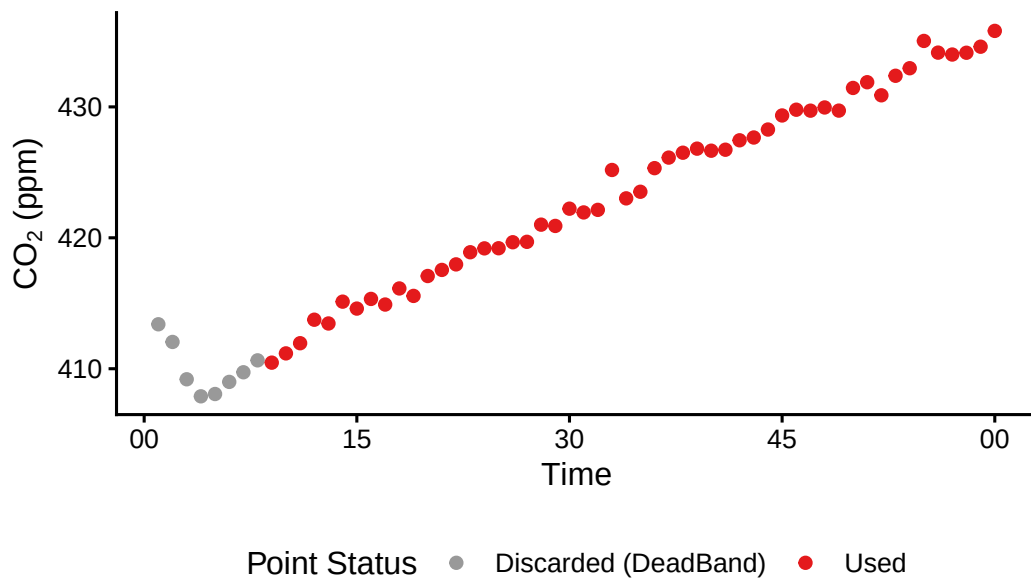
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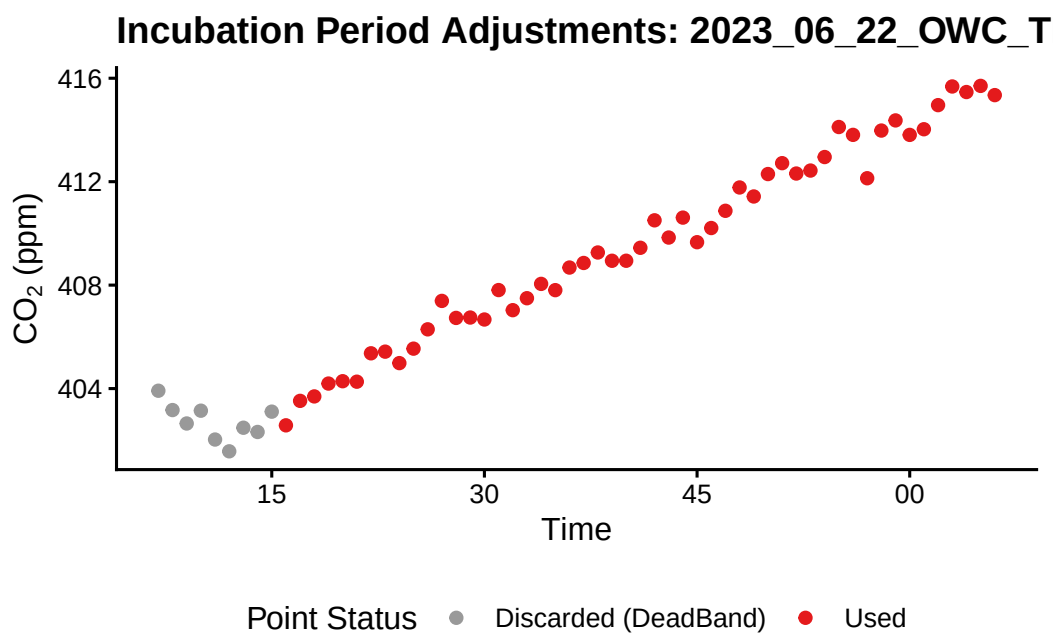
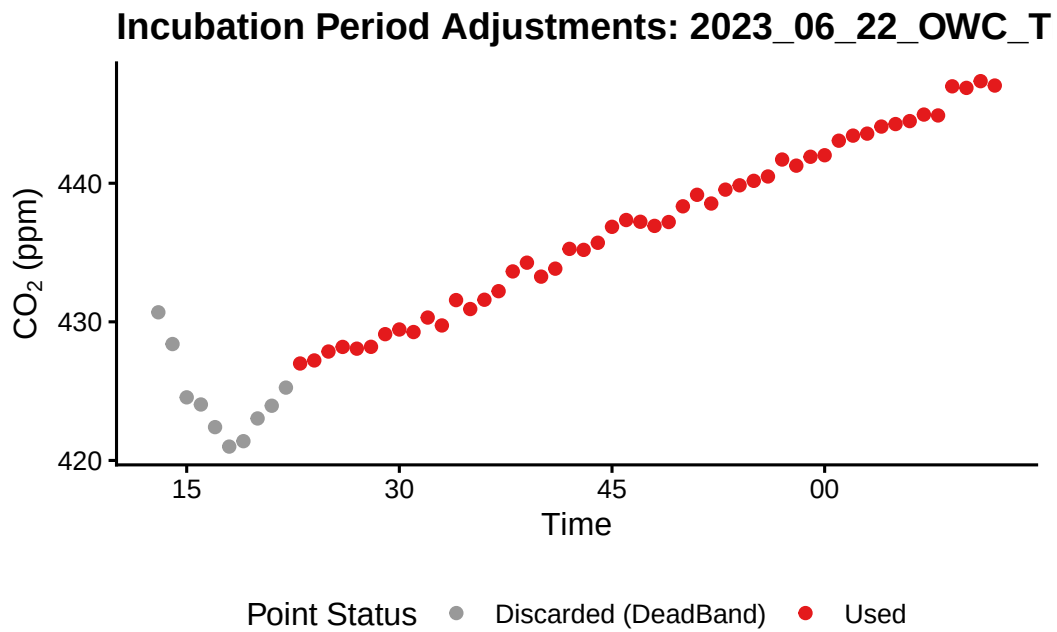


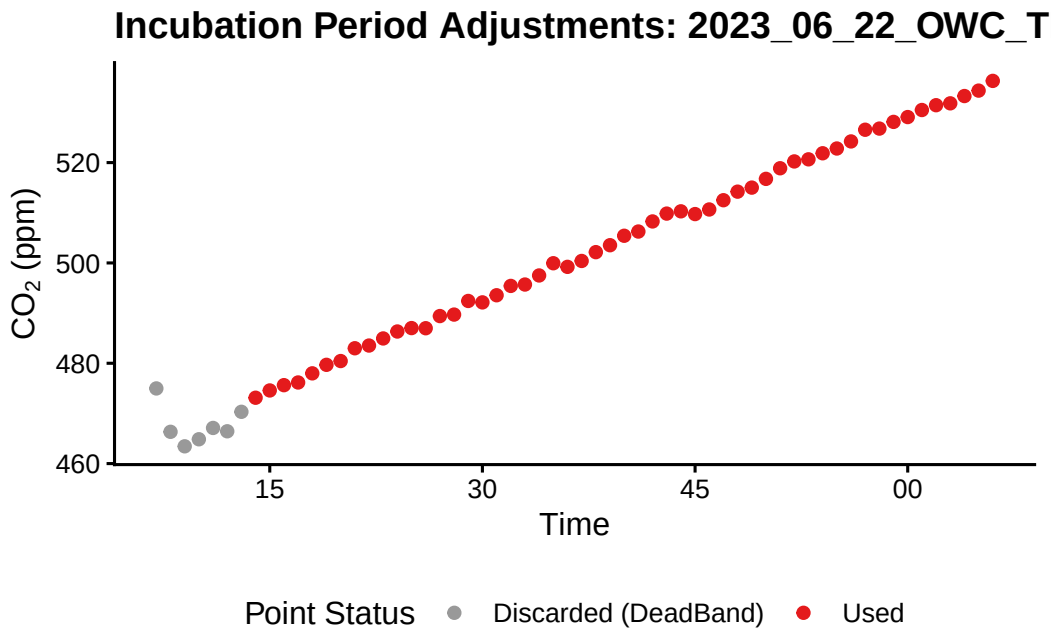
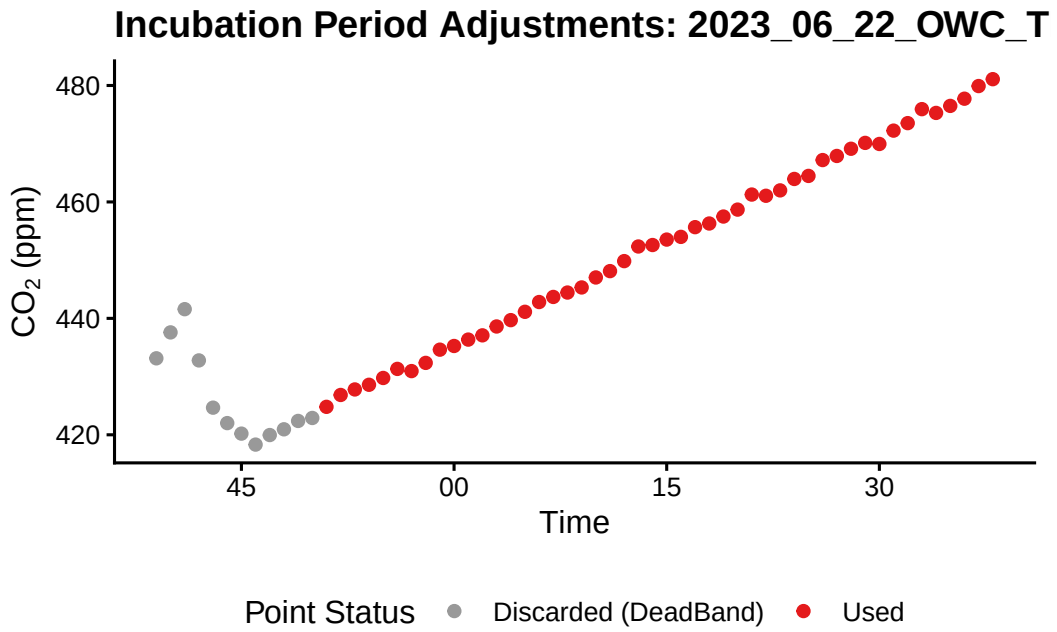
Incubation Period Adjustments: 2023_06_22_OWC_T



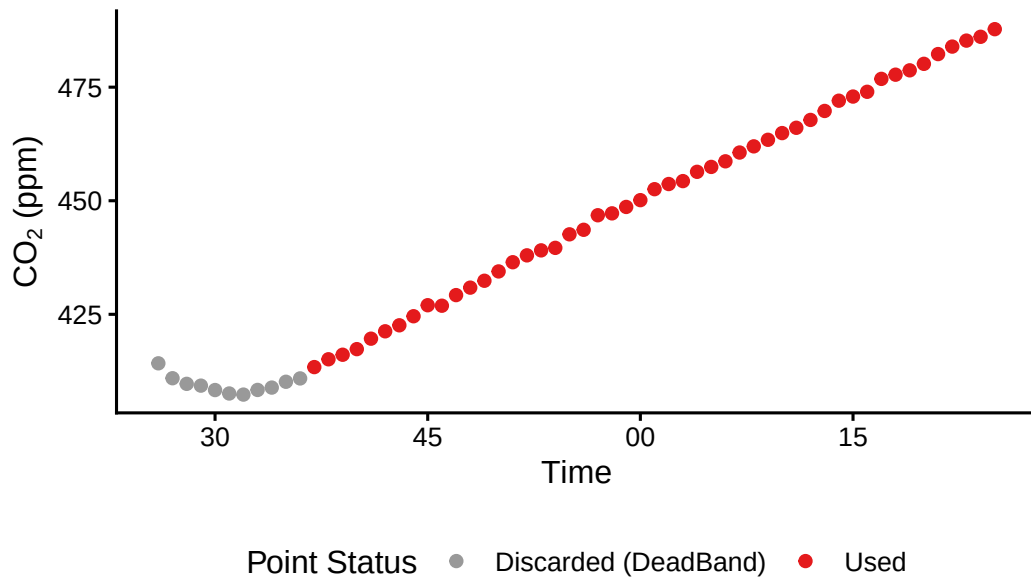
Incubation Period Adjustments: 2023_06_22_OWC_T



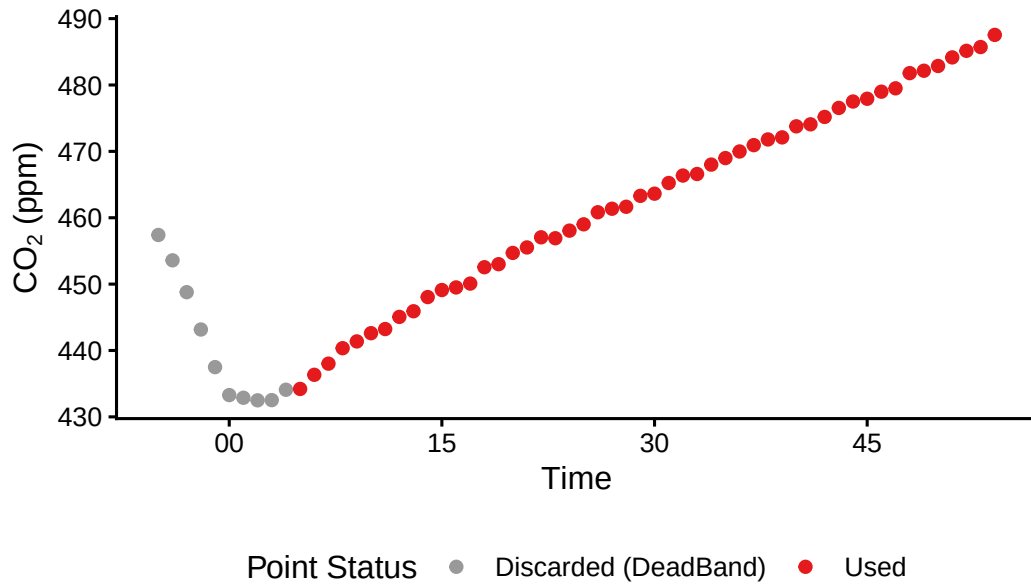


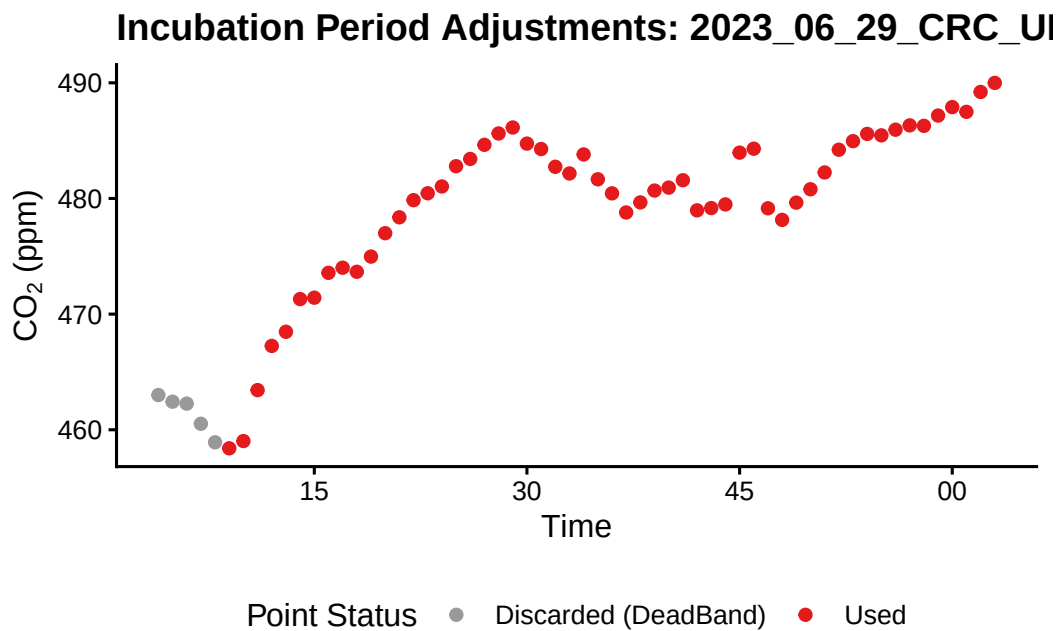
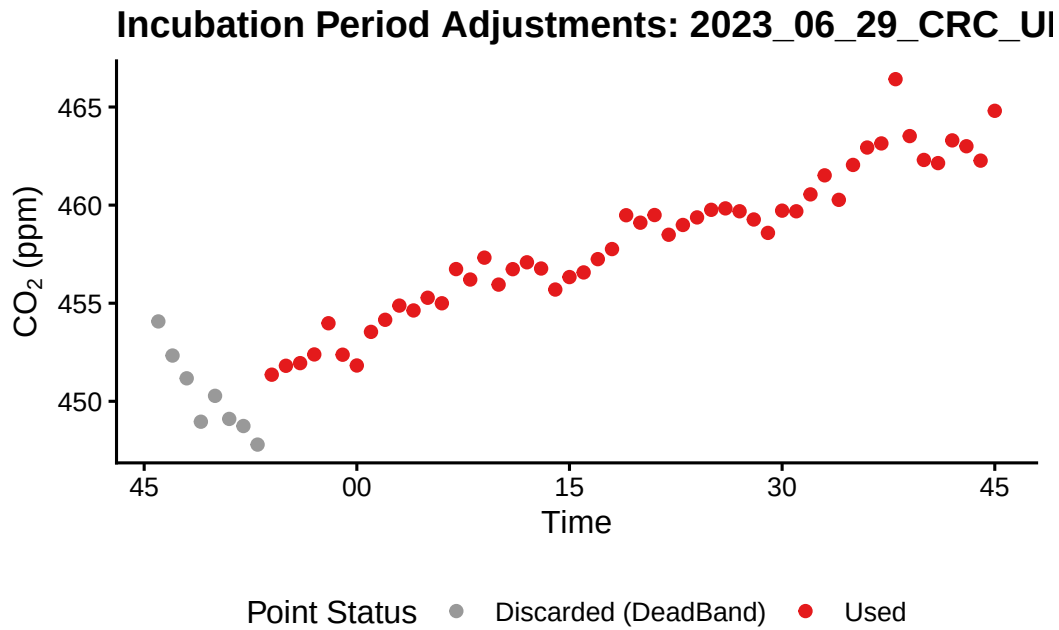


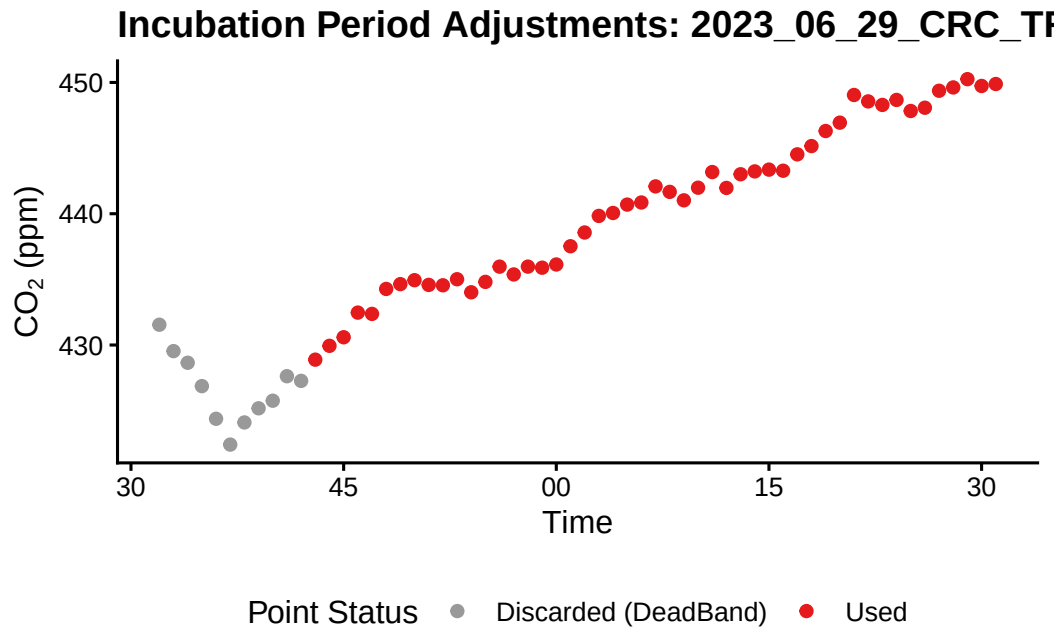
Incubation Period Adjustments: 2023_06_22_OWC_V



Incubation Period Adjustments: 2023_06_22_OWC_V

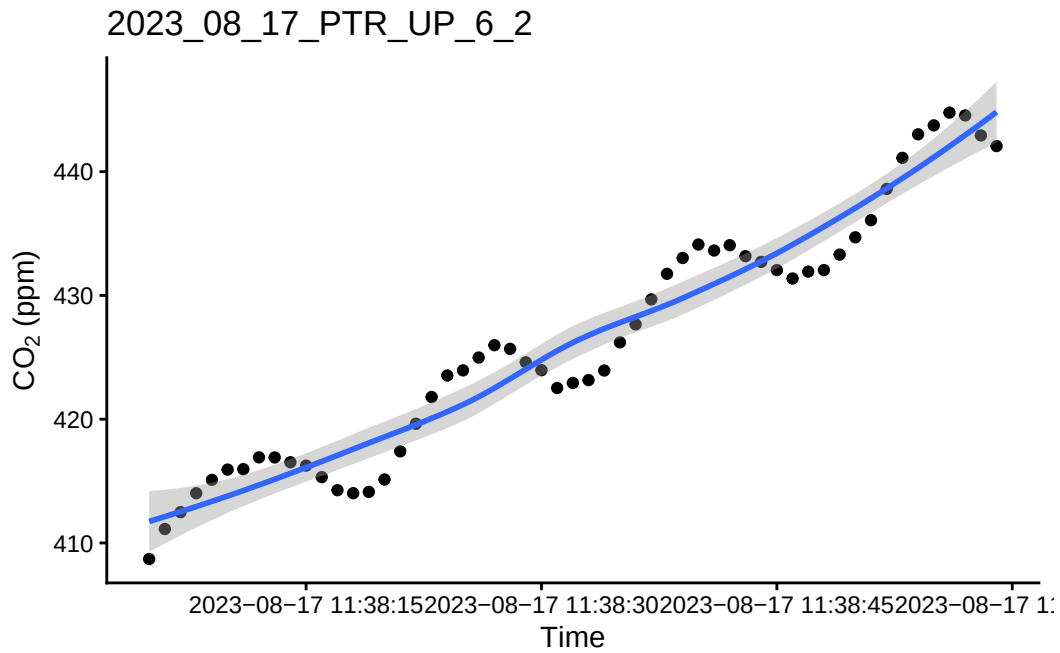




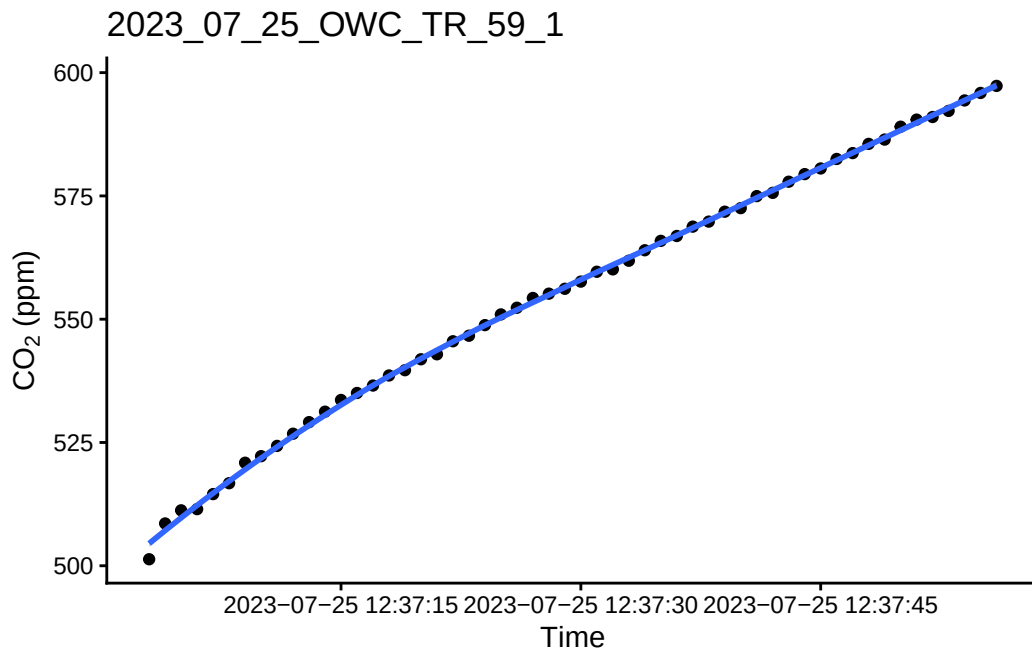


0.13 Spot check a few fluxes with $R^2 > 0.9$

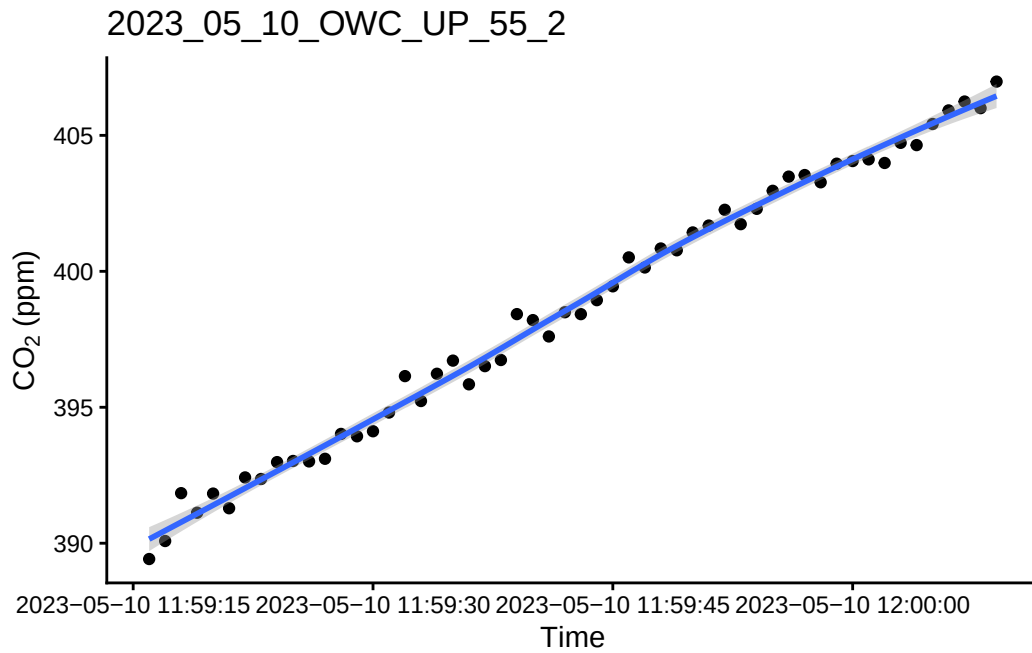
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



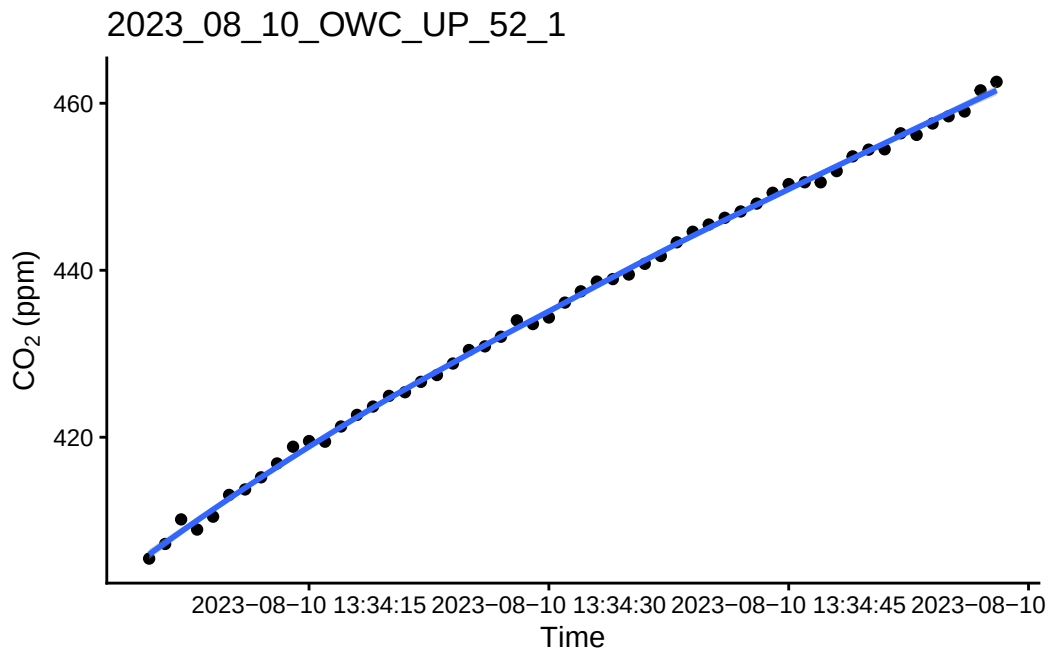
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



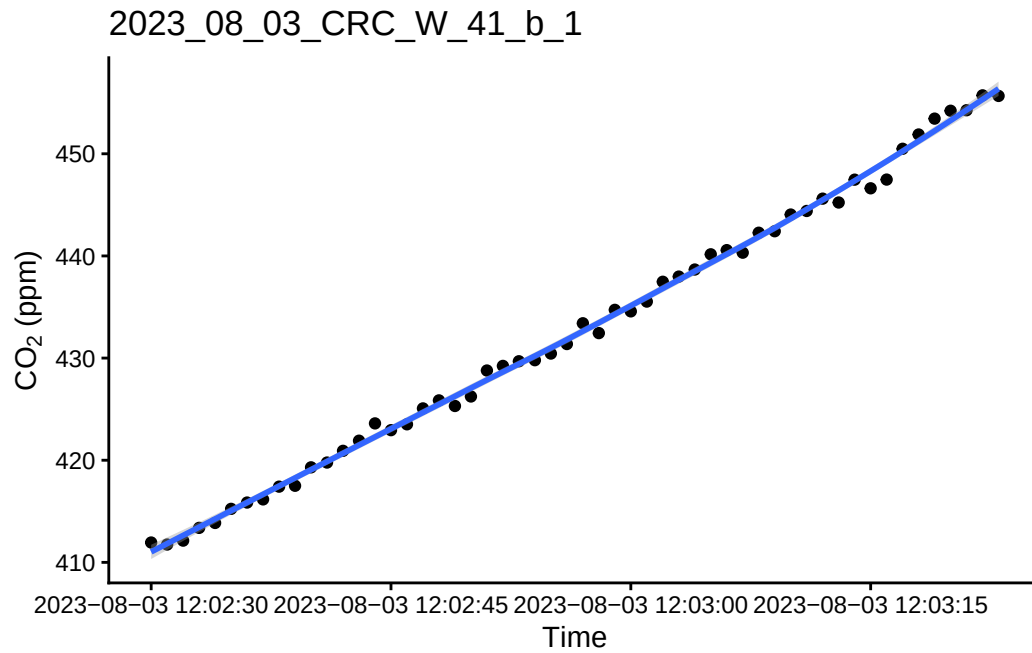
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

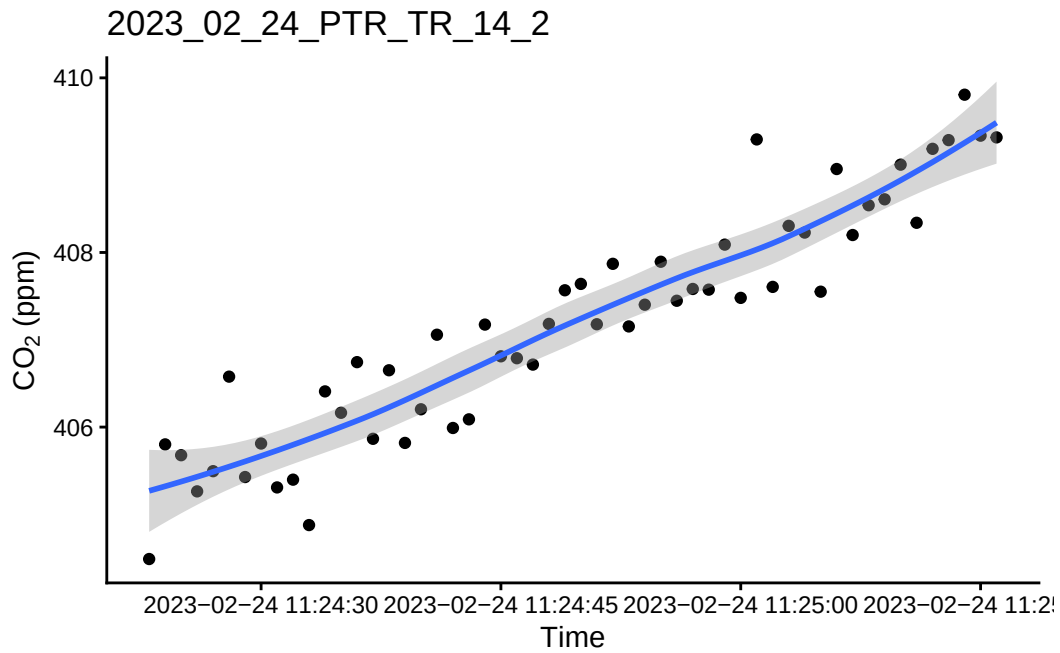


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

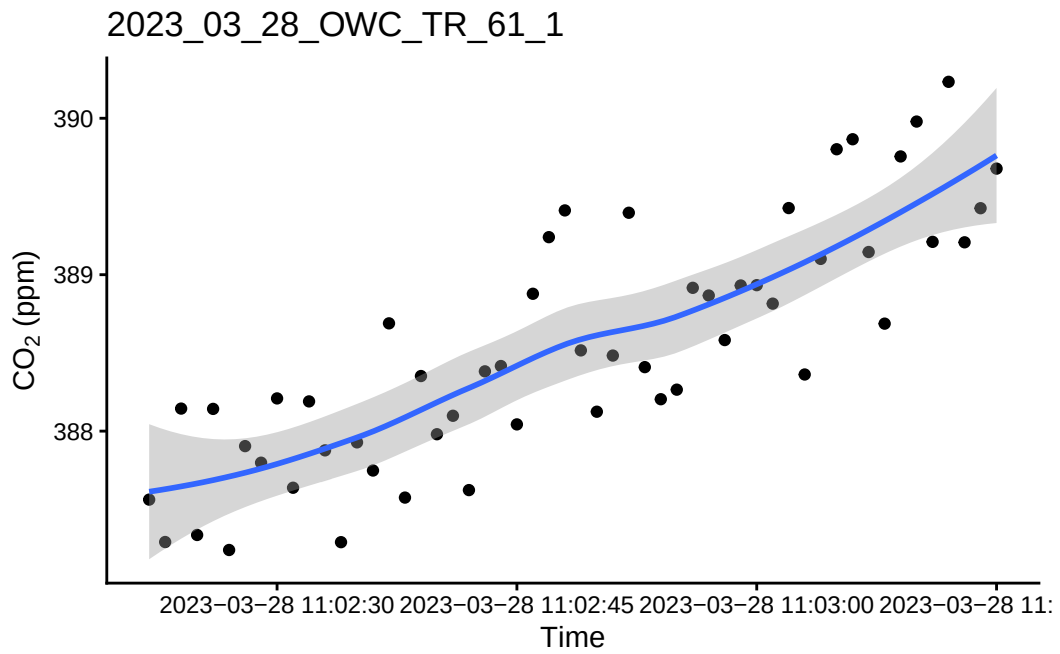


0.14 Take a look at the fluxes <0.9 and a flux estimate below 1

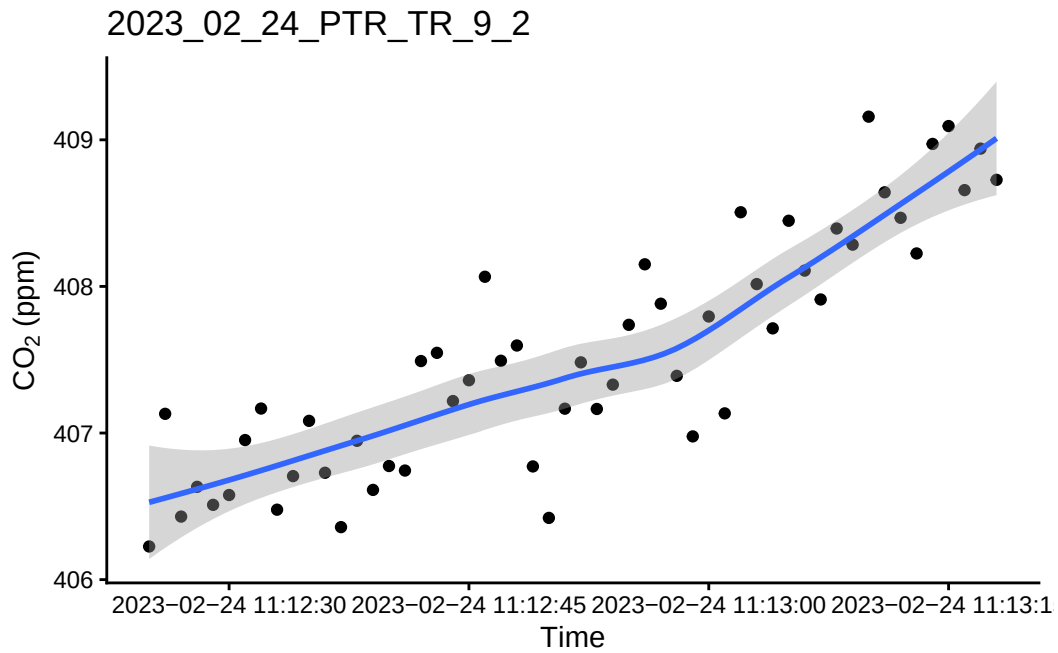
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



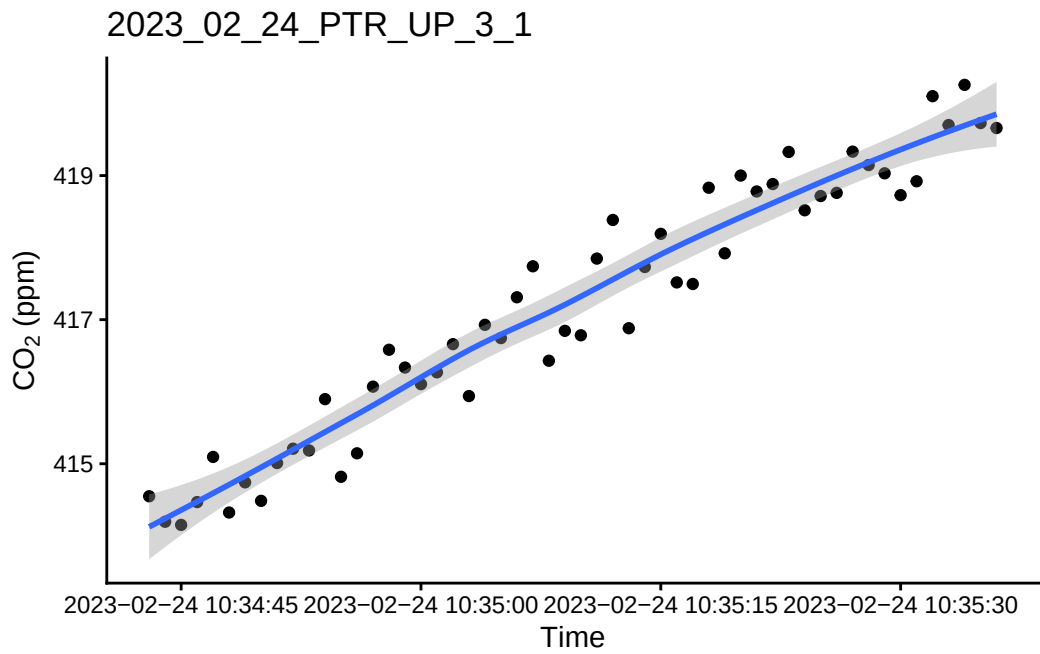
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



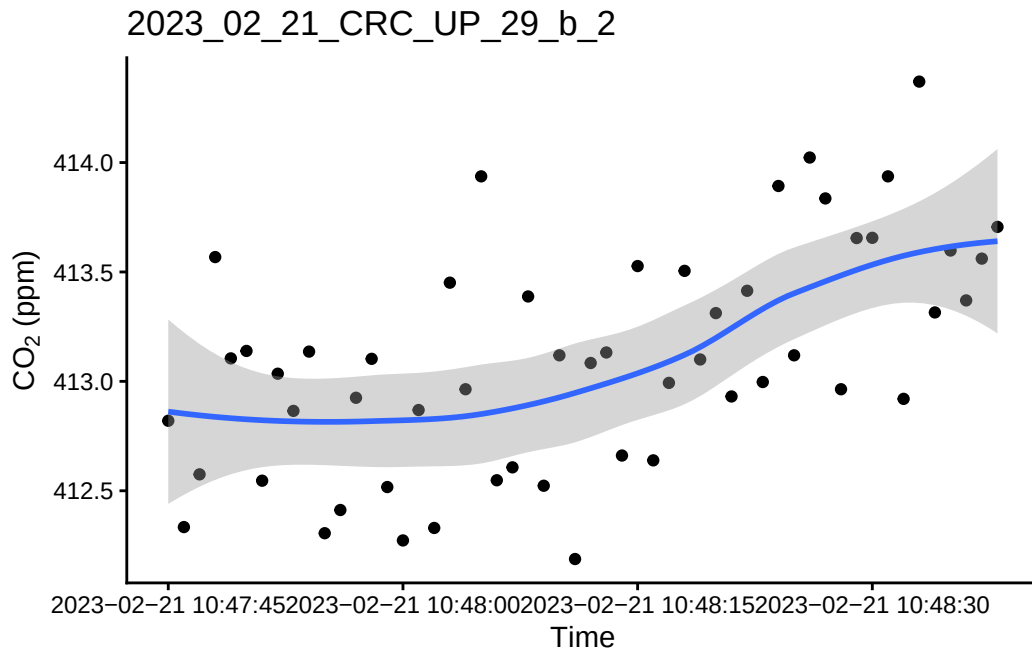
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



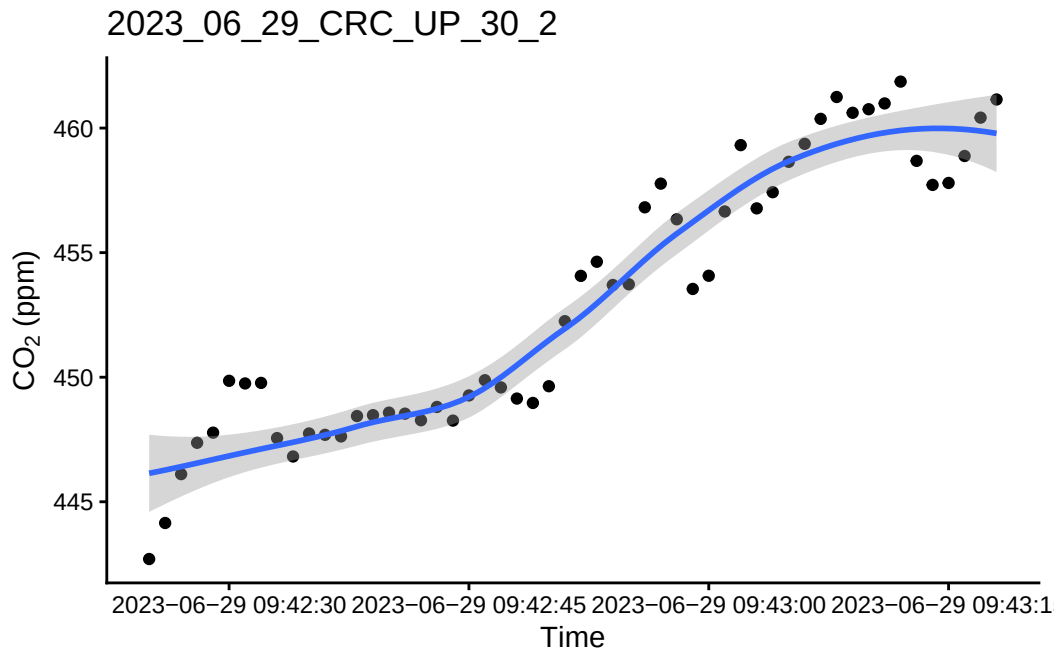
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



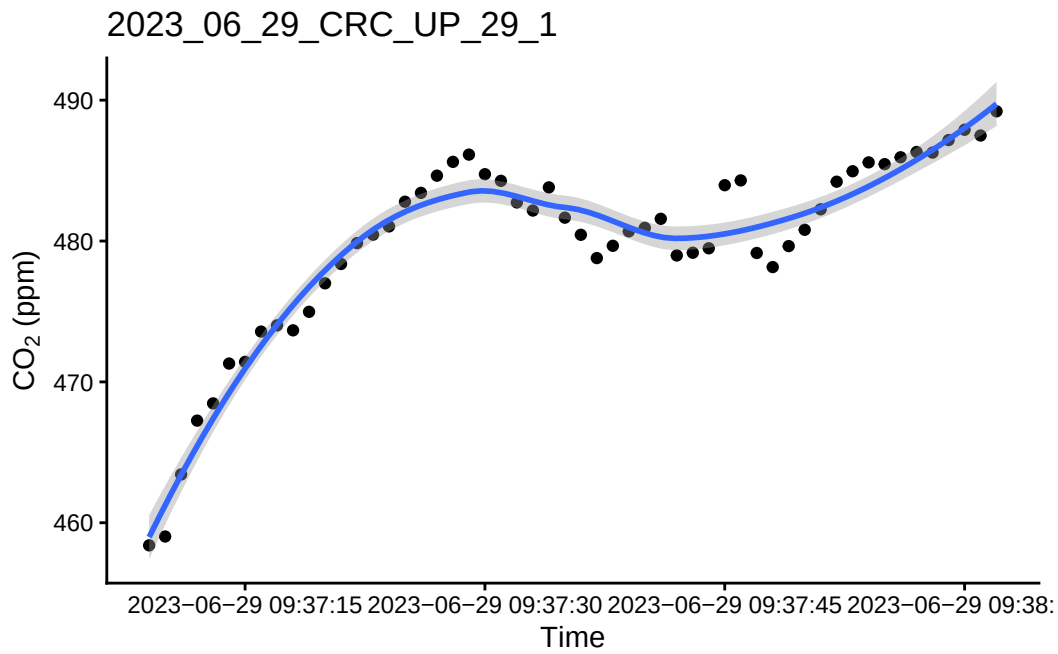
0.15 Take a look at the bad fluxes <0.9 and have a high flux estimate

1 See if we want to remove these fluxes or if we want to try and recalculate these ones?

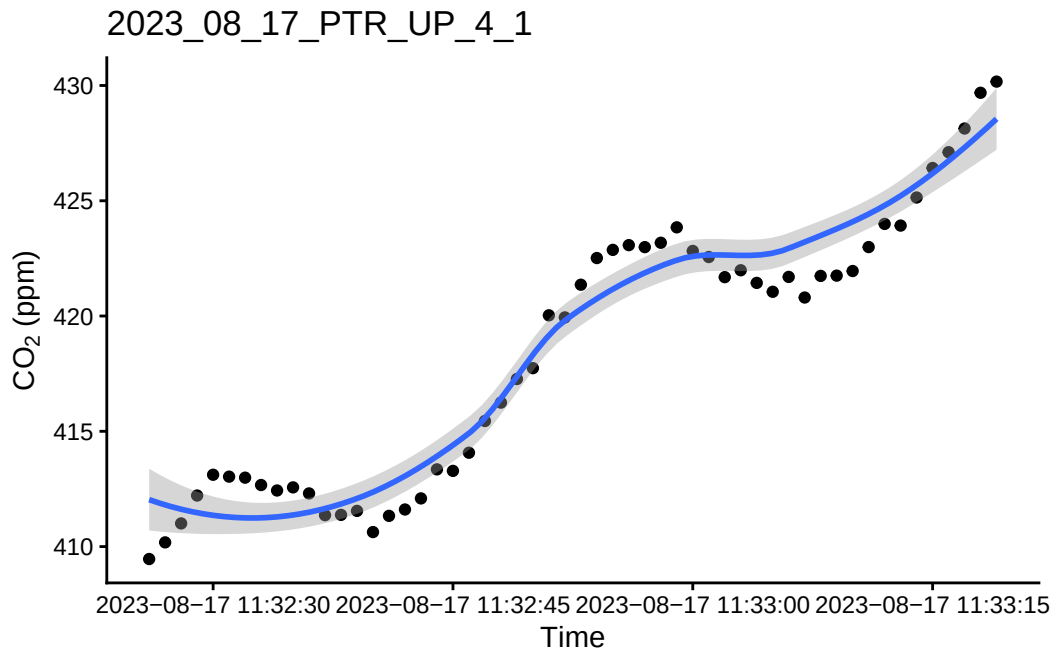
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



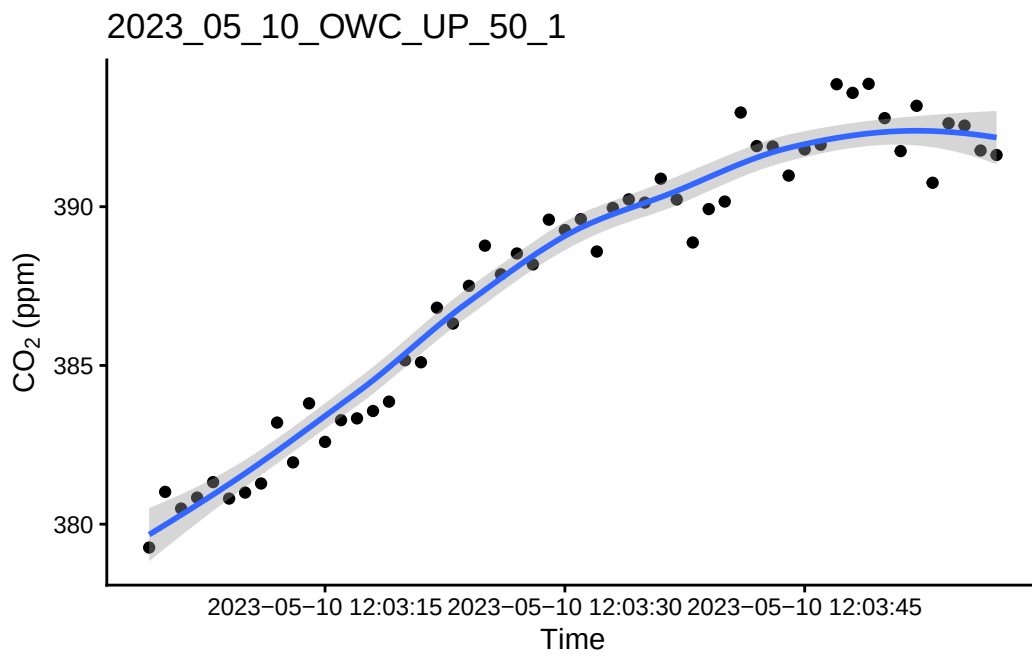
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



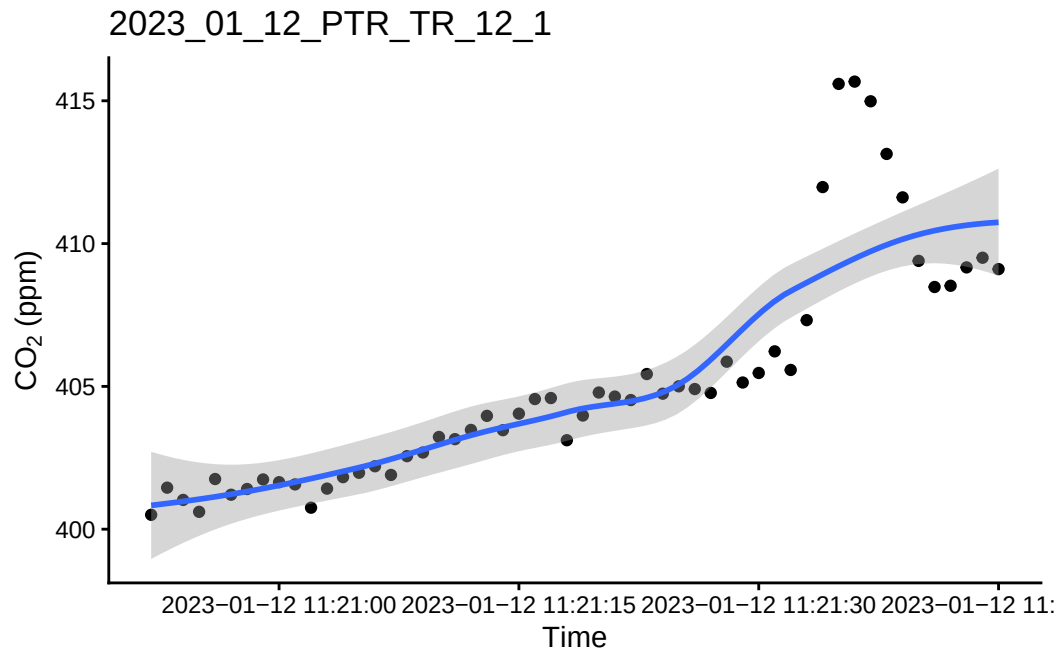
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

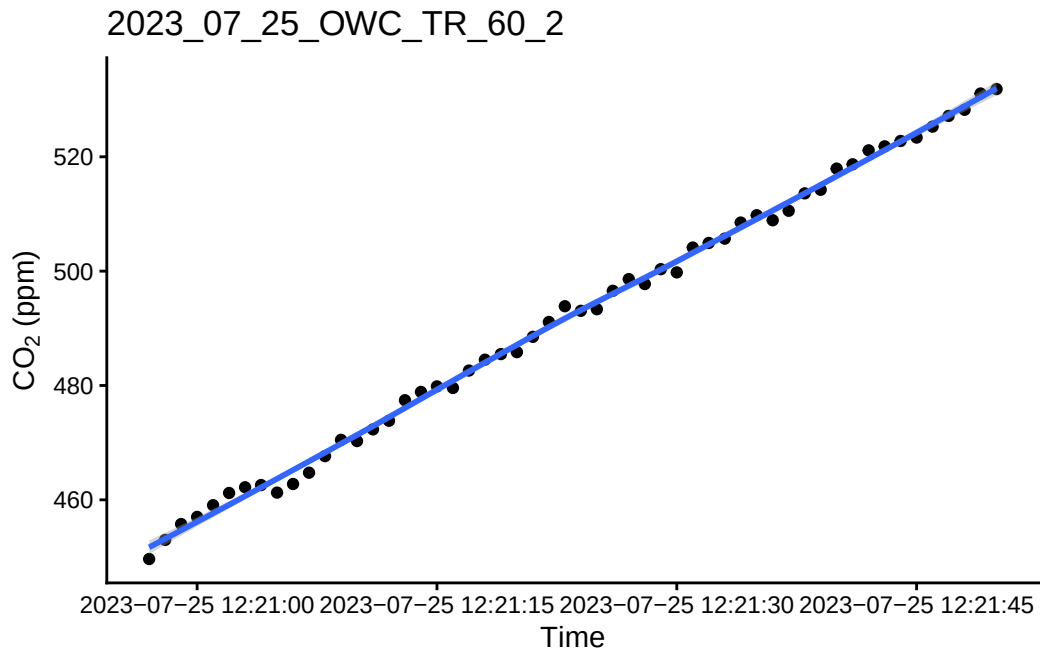


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

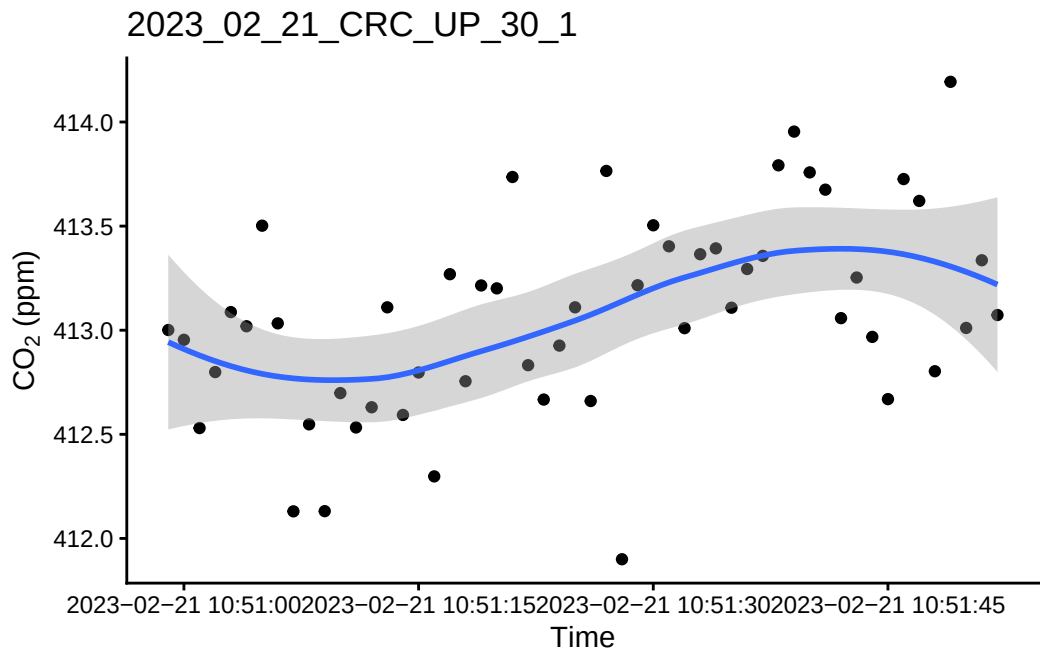


1.1 Look at the really high or really low ones to see if there is a timing issue or something

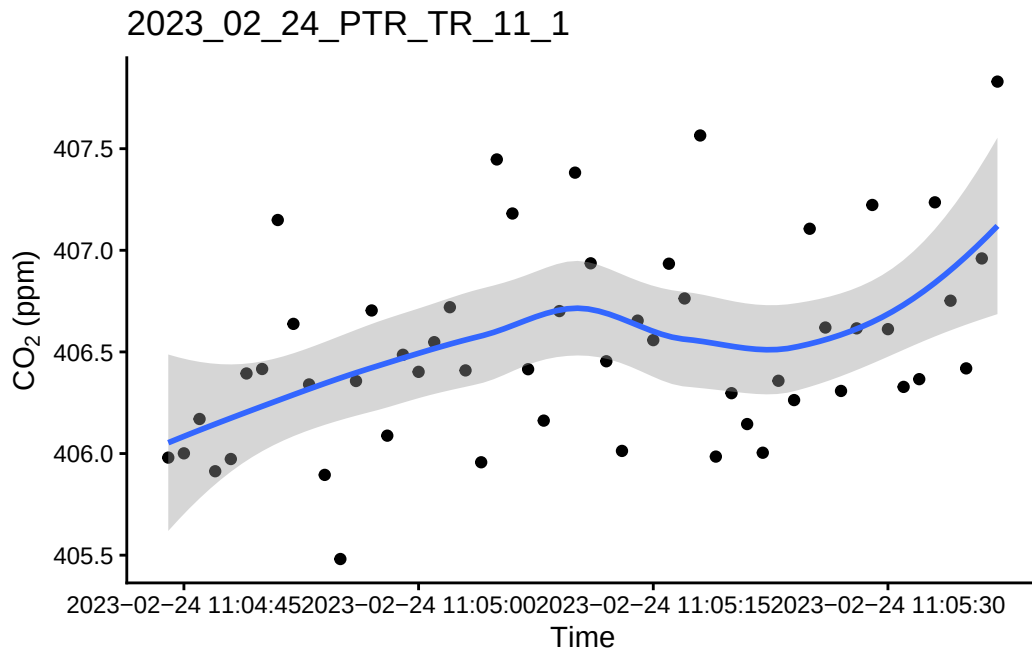
``geom_smooth()`` using `method = 'loess'` and `formula = 'y ~ x'`



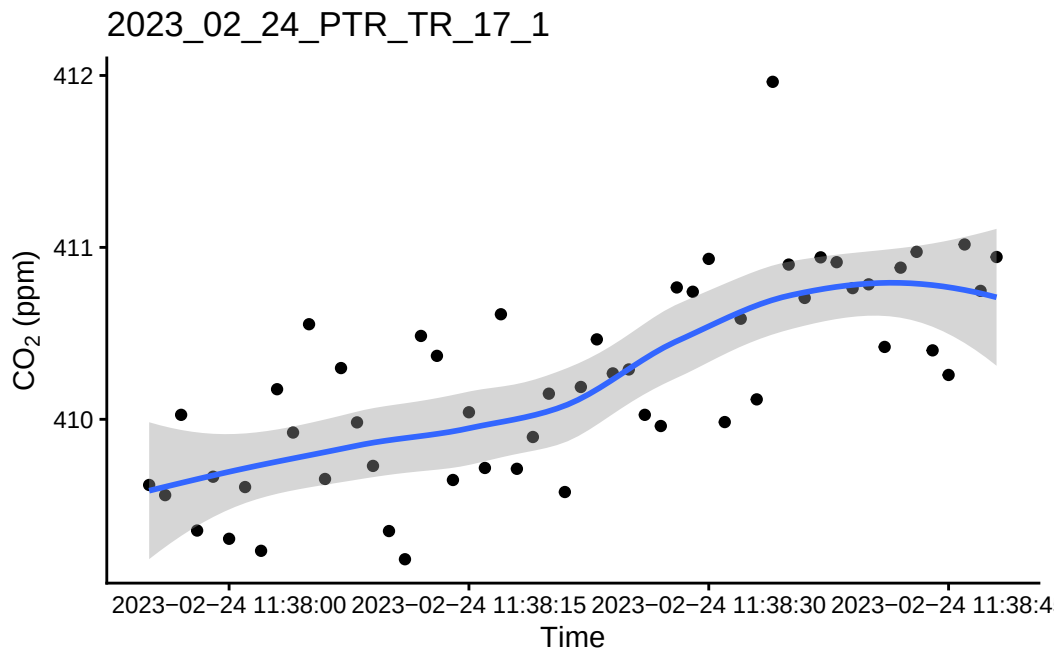
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



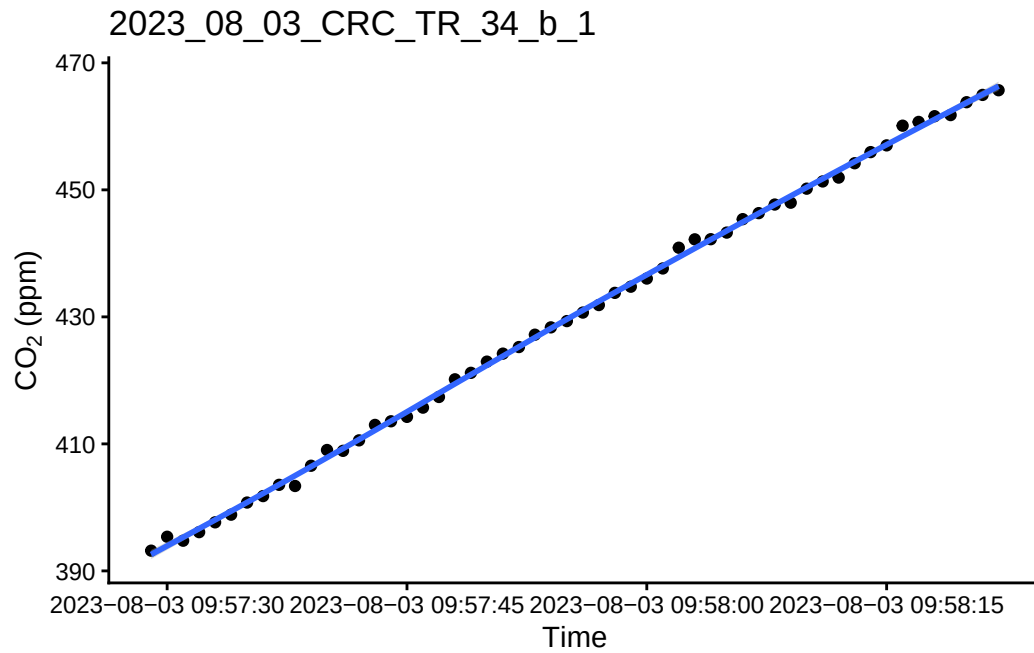
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

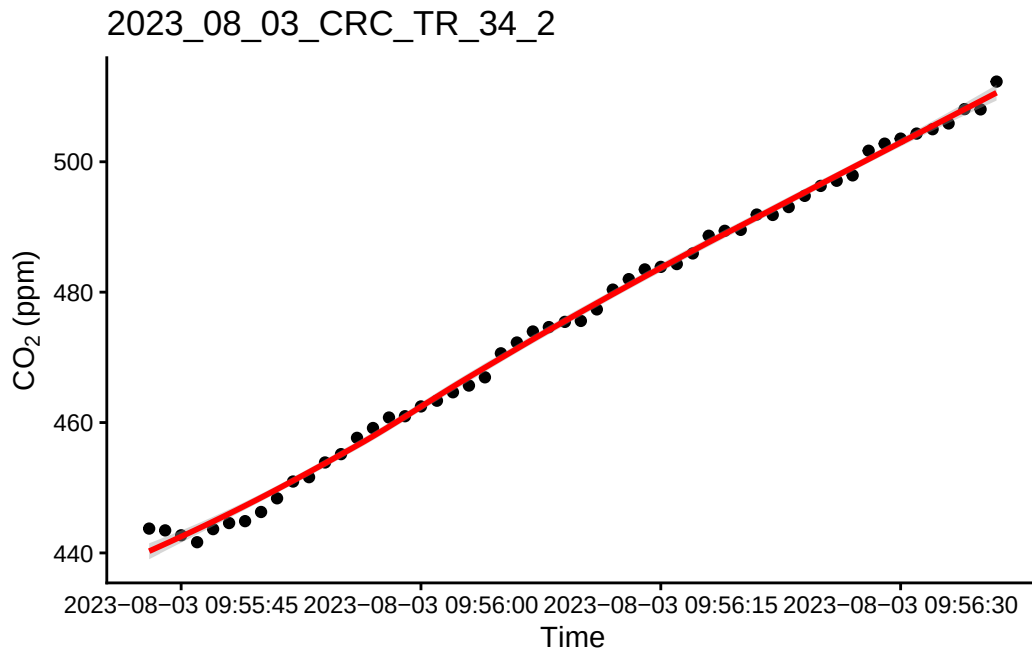


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

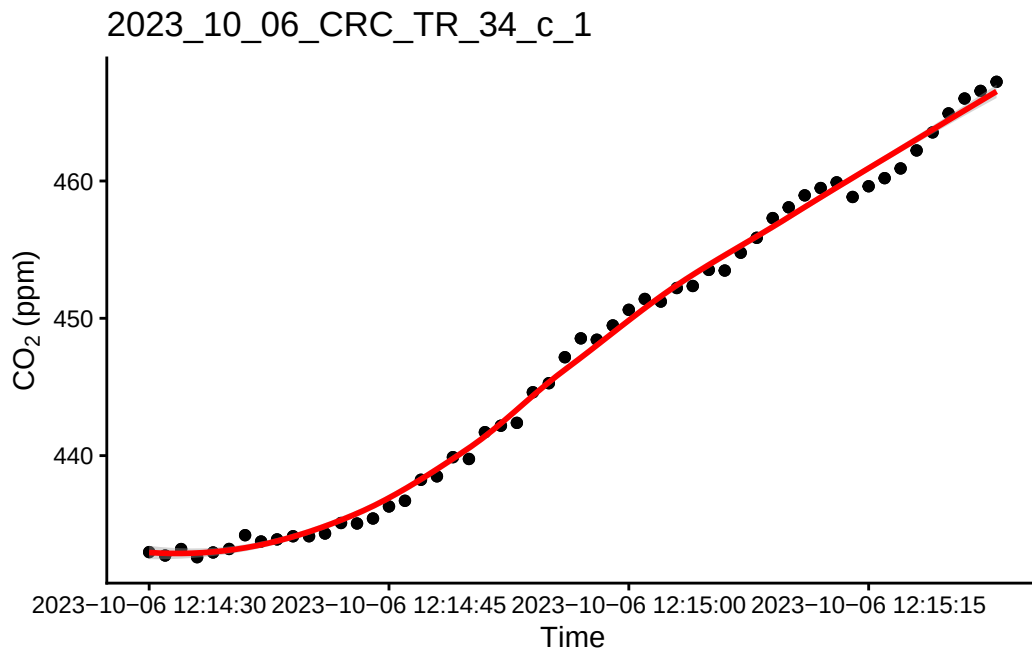


1.2 Check all fluxes

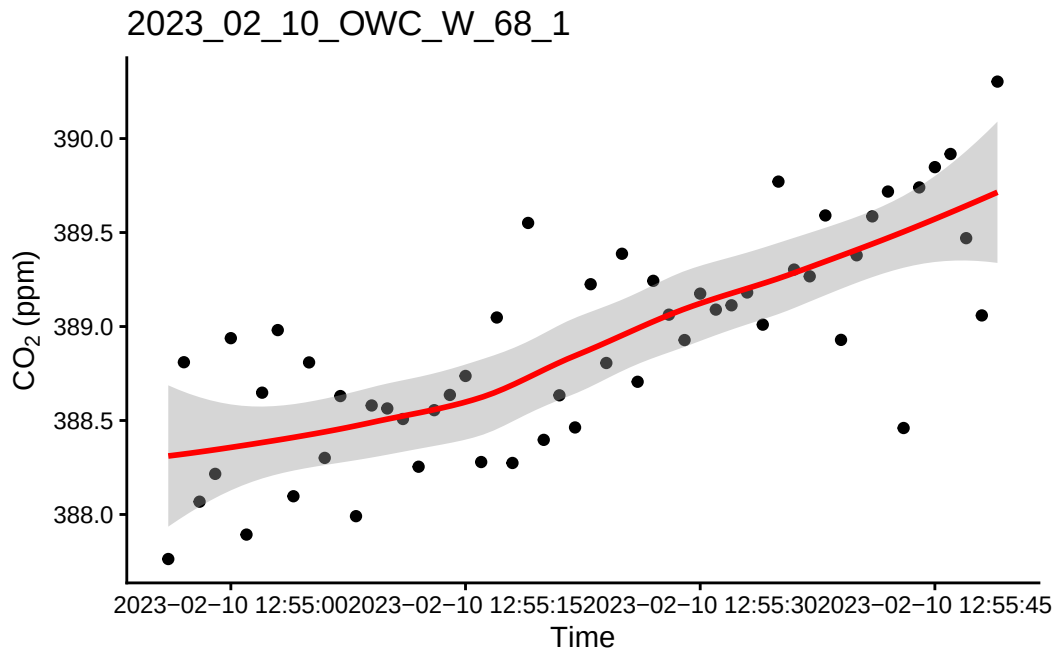
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



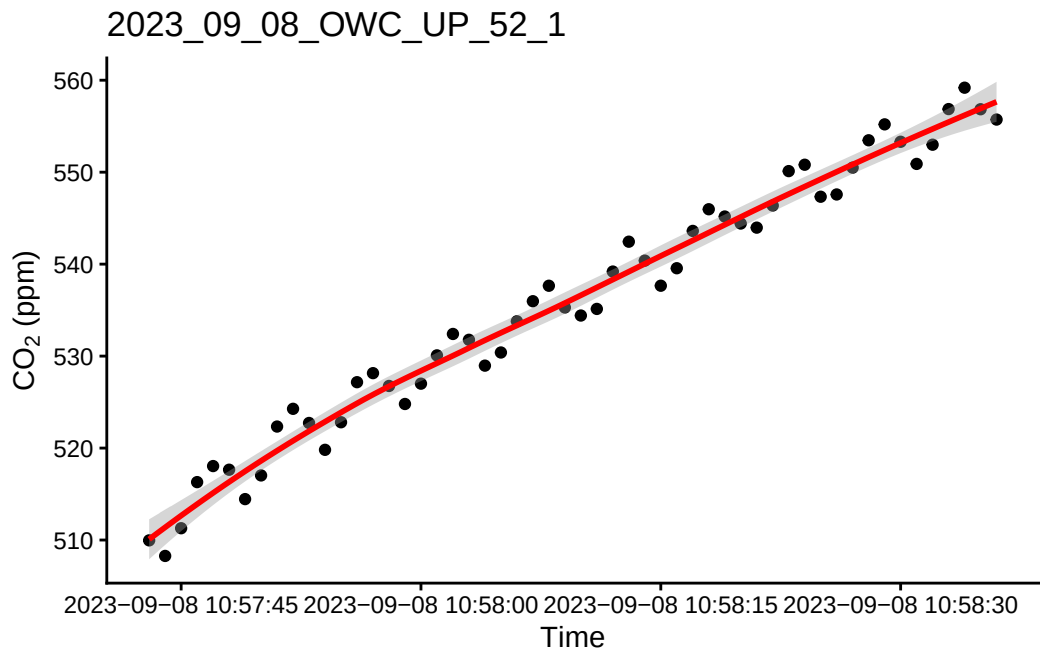
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



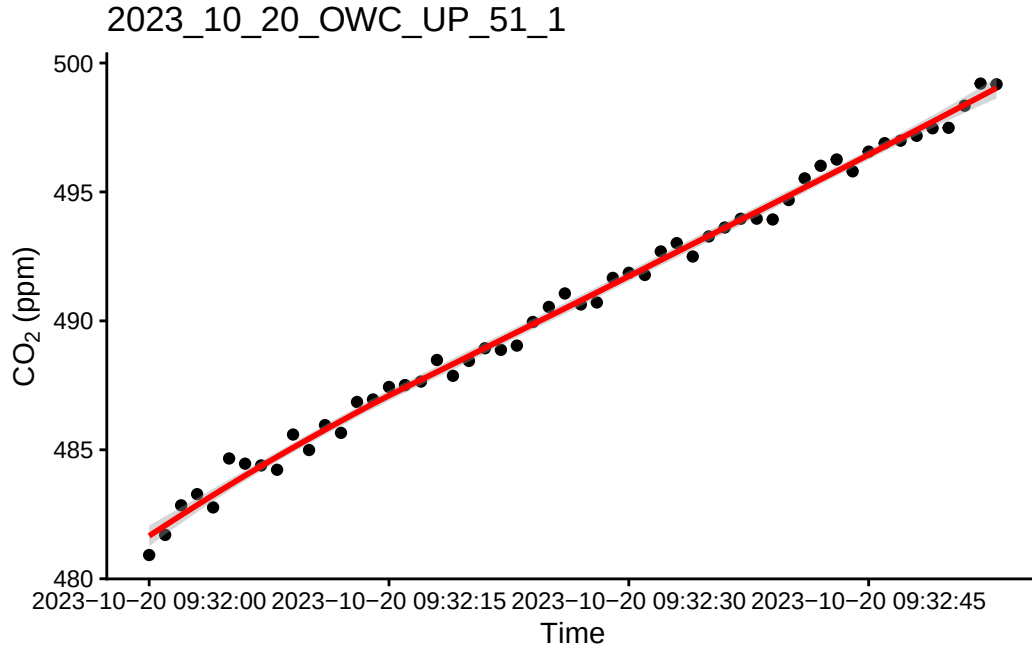
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

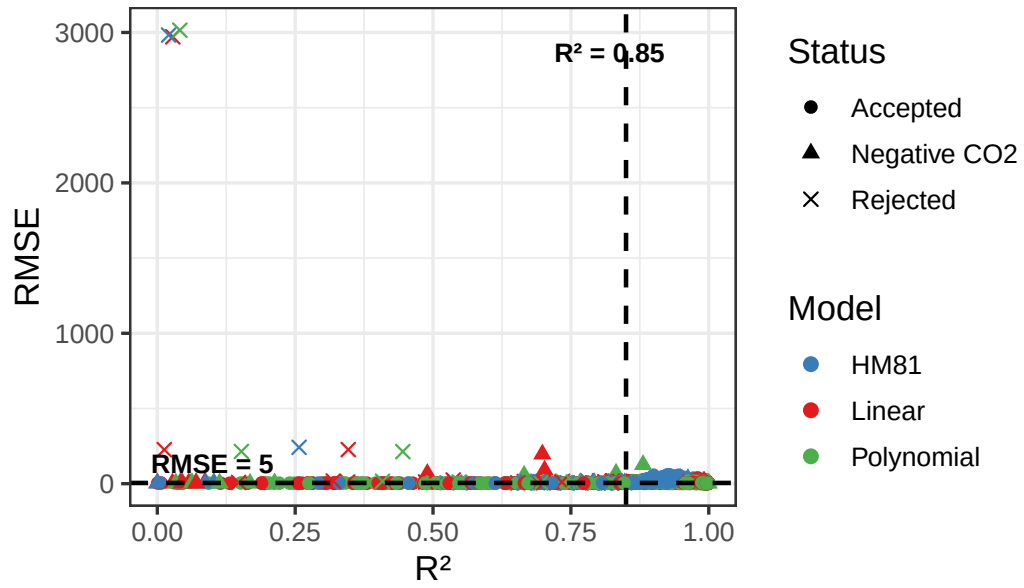


1.3 QAQC - Flags

Table 1: QA/QC CO₂ Flux Summary

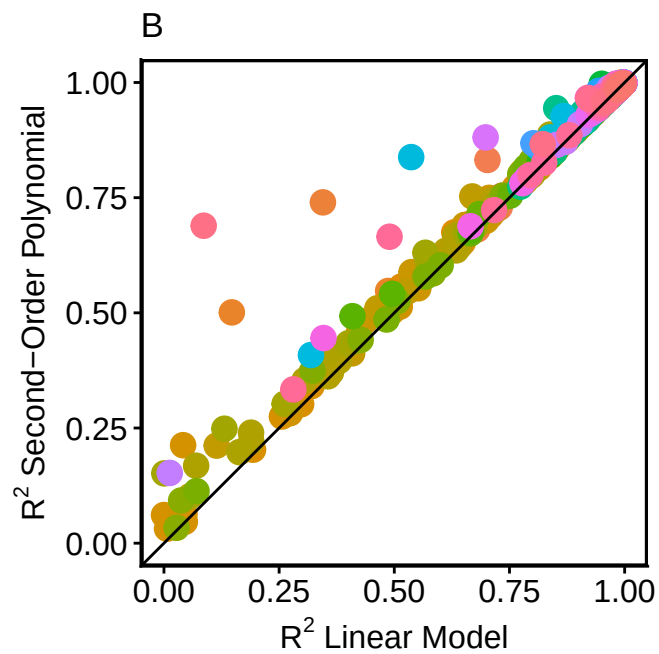
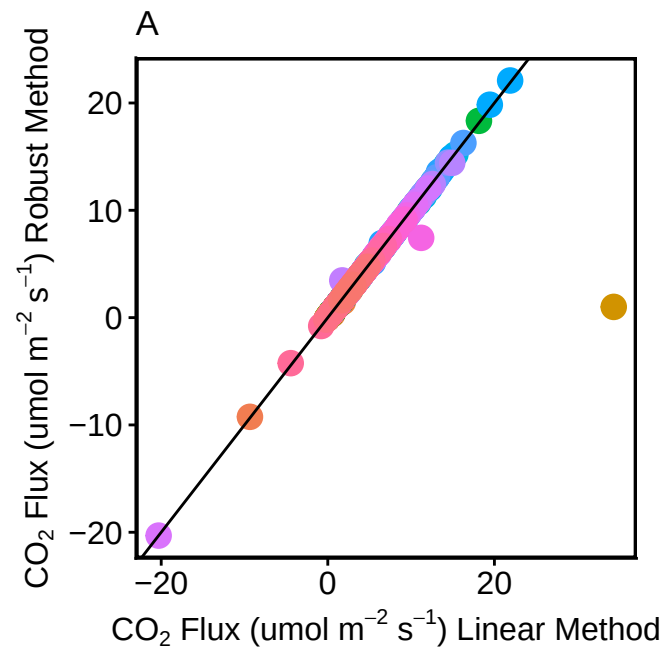
Metric	Value
Total observations	867.0
Rejections	
Rejected — poor model fit (n)	18.0
Rejected — negative CO ₂ flux (n)	14.0
Rejected — both reasons (n)	3.0
Rejected — total (n)	29.0
Rejected — total (%)	3.3
Rejected - [CO ₂] (n)	5.0
Rejected — [CO ₂] (%)	0.0
Model performance	
Passed — Linear model (n)	844.0
Passed — HM81 model (n)	476.0
Passed — Polynomial model (n)	848.0

R² vs RMSE — CO2 Flux QA/QC

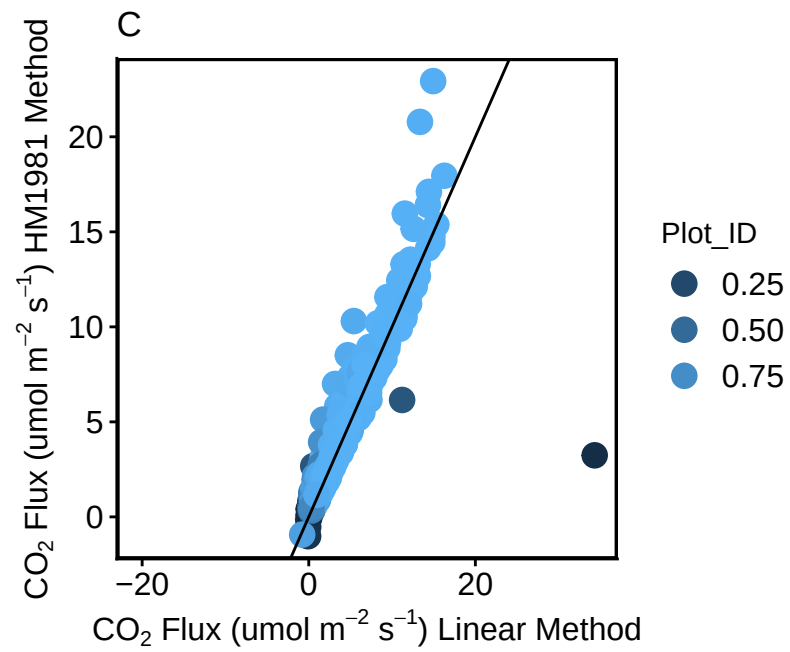


1.4 Adding manual flag to reject any fluxes that do not fit criteria

1.5 Do some visualization of QAQC plots to assess fluxes



Warning: Removed 370 rows containing missing values or values outside the scale range (``geom_point()``).



1.6 Read out final dataframe and export for use elsewhere

```
[1] "TR" "UP" "W"
```

```
[1] 58 60 61 63 64 50 51 52 55 56 65 66 68 71 72 11 12 14 15 9 2 3 4 6 7
[26] 17 18 20 22 23 34 36 37 38 39 25 26 29 41 42 43 46 47 48 62 67 27 30 8 33
[51] 53 45 70 35 40 59 32 69 28 1 21 44
```

```
[1] 1 2
```